

```
lab 3.1.py - C:\Users\ELCOT\AppData\Local\Programs\Python\Python313\lab 3.1.py (3.13.12)
File Edit Format Run Options Window Help
key = input("Enter key: ").upper()
text = input("Enter plaintext: ").upper()

# Create 5x5 matrix
alpha = "ABCDEFGHIKLMNOPQRSTUVWXYZ"
matrix = ""

for ch in key:
    if ch not in matrix and ch != "J":
        matrix += ch

for ch in alpha:
    if ch not in matrix:
        matrix += ch

print("Playfair Matrix:")
for i in range(0, 25, 5):
    print(matrix[i:i+5])

# Encrypt two letters
a = text[0]
b = text[1]

r1 = matrix.index(a) // 5
c1 = matrix.index(a) % 5
r2 = matrix.index(b) // 5
c2 = matrix.index(b) % 5

if r1 == r2: # Same row
    c1 = (c1 + 1) % 5
    c2 = (c2 + 1) % 5
elif c1 == c2: # Same column
    r1 = (r1 + 1) % 5
    r2 = (r2 + 1) % 5
else: # Rectangle rule
    c1, c2 = c2, c1

print("Cipher Text:", matrix[r1*5+c1] + matrix[r2*5+c2])

Ln: 1 Col: 0
```

```
IDLE Shell 3.13.12
File Edit Shell Debug Options Window Help
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb 3 2026, 18:22:25) [MSC v.1944 64
bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
== RESTART: C:\Users\ELCOT\AppData\Local\Programs\Python\Python313\lab 3.1.py ==
Enter key: vishal
Enter plaintext: he
Playfair Matrix:
VISHA
LBCDE
FGKMN
OPQRT
UWXYZ
Cipher Text: AD

>>>

Ln: 14 Col: 0
```

lab 4.1.py - C:\Users\ELCOT\AppData\Local\Programs\Python\Python313\lab 4.1.py (3.13.12)

File Edit Format Run Options Window Help

```
text = input("Enter text: ").upper()
key = input("Enter key: ").upper()

cipher = ""
j = 0

for i in text:
    if i.isalpha():
        x = (ord(i) - 65 + ord(key[j % len(key)]) - 65) % 26
        cipher = cipher + chr(x + 65)
        j += 1

print("Cipher Text:", cipher)
```

Ln: 1 Col: 0

IDLE Shell 3.13.12

File Edit Shell Debug Options Window Help

```
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb 3 2026, 18:22:25) [MSC v.1944 64
bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
== RESTART: C:\Users\ELCOT\AppData\Local\Programs\Python\Python313\lab 4.1.py
==
Enter text: vishal
Enter key: key
Cipher Text: FMQREJ

>>>
```

Ln: 8 Col: 0

lab 5.1.py - C:/Users/ELCOT/AppData/Local/Programs/Python/Python313/lab 5.1.py (3.13.12)

File Edit Format Run Options Window Help

```
text = input("Enter text: ").upper()
a = int(input("Enter a: "))
b = int(input("Enter b: "))

cipher = ""

for i in text:
    p = ord(i) - 65
    c = (a * p + b) % 26
    cipher = cipher + chr(c + 65)

print("Cipher Text:", cipher)
```

Ln: 13 Col: 0

IDLE Shell 3.13.12

File Edit Shell Debug Options Window Help

```
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb 3 2026, 18:22:25) [MSC v.1944 64
bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

== RESTART: C:/Users/ELCOT/AppData/Local/Programs/Python/Python313/lab 5.1.py
==
Enter text: vishal
Enter a: 5
Enter b: 8
Cipher Text: JWURIL

>>>
```

Ln: 9 Col: 0

```
lab 6.1.py - C:/Users/ELCOT/AppData/Local/Programs/Python/Python313/lab 6.1.py (3.13.12)
File Edit Format Run Options Window Help
cipher = input("Enter cipher text: ").upper()

for a in [1,3,5,7,9,11,15,17,19,21,23,25]:
    for b in range(26):
        plain = ""

        for ch in cipher:
            if ch.isalpha():
                for x in range(26):
                    if (a * x + b) % 26 == ord(ch) - 65:
                        plain += chr(x + 65)
                        break
            else:
                plain += ch

        print("a =", a, "b =", b, ":", plain)
```

Ln: 17 Col: 0

```
IDLE Shell 3.13.12
File Edit Shell Debug Options Window Help
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb 3 2026, 18:22:25) [MSC v.1944 64
bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
== RESTART: C:/Users/ELCOT/AppData/Local/Programs/Python/Python313/lab 6.1.py ==
Enter cipher text: BUB
a = 1 b = 0 : BUB
a = 1 b = 1 : ATA
a = 1 b = 2 : ZSZ
a = 1 b = 3 : YRY
a = 1 b = 4 : XQX
a = 1 b = 5 : WPW
a = 1 b = 6 : VOV
a = 1 b = 7 : UNU
a = 1 b = 8 : TMT
a = 1 b = 9 : SLS
a = 1 b = 10 : RKR
a = 1 b = 11 : QJQ
a = 1 b = 12 : PIP
a = 1 b = 13 : OHO
a = 1 b = 14 : NGN
a = 1 b = 15 : MFM
a = 1 b = 16 : LEL
a = 1 b = 17 : KDK
a = 1 b = 18 : JCJ
a = 1 b = 19 : IBI
a = 1 b = 20 : HAH
a = 1 b = 21 : GZG
a = 1 b = 22 : FYF
a = 1 b = 23 : EXE
a = 1 b = 24 : DWD
a = 1 b = 25 : CVC
a = 3 b = 0 : JYJ
a = 3 b = 1 : APA
a = 3 b = 2 : RGR
a = 3 b = 3 : IXI
a = 3 b = 4 : ZOZ
a = 3 b = 5 : QFQ
a = 3 b = 6 : HWH
```

Ln: 318 Col: 0

lab 2.1py.py - C:\Users\ELCOT\AppData\Local\Programs\Python\Python313\lab 2.1py.py (3.13.12)

File Edit Format Run Options Window Help

```
import string

alphabet = string.ascii_uppercase

key = input("Enter the 26-letter substitution key: ").upper()

if len(key) != 26 or set(key) != set(alphabet):
    print("Invalid key! Please enter all 26 unique letters.")
    exit()

plaintext = input("Enter the plaintext: ").upper()

ciphertext = ""

for ch in plaintext:
    if ch in alphabet:
        index = alphabet.index(ch)
        ciphertext += key[index]
    else:
        ciphertext += ch

print("Plaintext :", plaintext)
print("Ciphertext:", ciphertext)
```

Ln: 1 Col: 0

IDLE Shell 3.13.12

File Edit Shell Debug Options Window Help

```
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb 3 2026, 18:22:25) [MSC v.1944 64
bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:\Users\ELCOT\AppData\Local\Programs\Python\Python313\lab 2.1py.p
y =
Enter the 26-letter substitution key: mnbvcxzasdfghjklpoiuytrewq
Enter the plaintext: vishal
Plaintext : VISHAL
Ciphertext: TSIAMG

>>>
```

Ln: 9 Col: 0

```
lab 1.1.py - C:\Users\ELCOT\AppData\Local\Programs\Python\Python313\lab 1.1.py (3.13.12)
File Edit Format Run Options Window Help

k = int(input("Key: "))
t = input("Text: ")

r = ""
for c in t:
    if c.isalpha():
        b = 65 if c.isupper() else 97
        r += chr((ord(c) - b + k) % 26 + b)
    else:
        r += c
print(r)

Ln: 1 Col: 0
```

```
IDLE Shell 3.13.12
File Edit Shell Debug Options Window Help

Python 3.13.12 (tags/v3.13.12:1cbe481, Feb 3 2026, 18:22:25) [MSC v.1944 64
bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
== RESTART: C:\Users\ELCOT\AppData\Local\Programs\Python\Python313\lab 1.1.py ==
Key: 4
Text: vishal
zmvlep

>>>

Ln: 8 Col: 0
```